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Provenance Analysis Based on Cluster In-Betweenness and Support Vector Machines: Identifying Migrant Candidates Using Multi-Isotope Fingerprints

Andrea Göhring[1,2] | Daniyal Kazempour[2] | Jan Robert Bensien[2] | Sweety Mohanty[2] | Claudius Zelenka[2] | Matthias Renz[2] | Peer Kröger[2] | Gisela Grupe[3] | Amei Lang[4]

[1]Leibniz-Laboratory for Radiometric Dating and Stable Isotope Research, Kiel University, Kiel, Germany | [2]Faculty of Engineering, Department of Computer Science, Kiel University, Kiel, Germany | [3]Faculty of Biology, Department of Biology I, Anthropology and Environmental History, Ludwig Maximilian University Munich, Planegg-Martinsried, Germany | [4]Faculty of Cultural Studies, Department for Cultural Sciences and Ancient Studies, Institute for Prehistory and Early Historic Archaeology and the Archaeology of the Roman Provinces, Ludwig Maximilian University Munich, Munich, Germany

Correspondence: Andrea Göhring (agoehring@leibniz.uni-kiel.de)

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ABSTRACT

Provenance reconstruction using strontium and lead stable isotopes can produce complex multidimensional fingerprints, challenging traditional methods. Identifying nonlocals, who migrated between sites, is a major task. Migrants are identifiable by divergent multi-isotope fingerprints due to isotopic mixing between origin and destination sites. Detecting early migrants, however, is possible in exceptional cases only.

This study used a Support Vector Machine approach to detect migrants as "in-between instances" between sites. Tested on synthetic data, the model effectively identified IBIs under varying conditions. Applied to archaeological multi-isotope data, it revealed nonlocal individuals, also early migrants, including the approximated years since migration.

1 | Introduction

1.1 | Multi-Isotope Fingerprints and Machine Learning

The analysis of multi-isotope fingerprints has been commonly used since the last decade and clearly represents an important tool in the field of bioarchaeology. Multidimensional isotope signatures, characteristic for an individual, dietary habits, and/or a region, allow to fingerprint the diet, water sources, habitat, or origin of an individual (Grupe et al. 2017; Mauder et al. 2017). Multi-isotope data including strontium ($^{87}Sr/^{86}Sr$) and lead stable isotope ratios ($^{208}Pb/^{204}Pb$, $^{207}Pb/^{204}Pb$, $^{206}Pb/^{204}Pb$, $^{208}Pb/^{207}Pb$, $^{206}Pb/^{207}Pb$) can be used to, e.g., reconstruct the provenance of individuals or identify herding strategies (e.g., Balasse et al. 2002; Boldsen et al. 2024; Burton and Price 2013; Göhring et al. 2023; Grupe et al. 2020; Kamenov and Krigbaum 2023; Sabaux et al. 2024; Schjerven et al. 2024; Tschetsch et al. 2020). Bioavailable strontium primarily reflects the underlying geology of a region, the soil composition, and precipitation (Lengfelder et al. 2019). Lead isotopes can be used to, e.g., identify mining areas or ore deposits, thus identifying (non)local individuals at a site or in a region (Price et al. 2017). In the case of inhumated remains, isotope analysis is mainly conducted on enamel (preferably on the first permanent molar), but also on bones. The analysis of both enamel and bone samples of the same individual

Andrea Göhring and Daniyal Kazempour shared first authorship.

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allows to study intra-life migration (see also below; e.g., Grupe et al. 1997; Schweissing and Grupe 2003; Göhring et al. 2020). It has, however, to be kept in mind that bones are generally more prone to diagenesis and contamination by the surrounding soil. This can be checked using, e.g., mineralogical analyses (e.g., Schmahl et al. 2017). In the case of cremated remains, calcined bones are used (e.g., Grupe et al. 2021; Harvig et al. 2014; Snoeck et al. 2022). The latter are usually more robust against contamination.

Isotope ratios in skeletal tissue are also influenced by the remodeling rate of the analyzed sample material. Enamel reflects the isotope signal of the period of formation, during (early) childhood or, in case of the third molar, up to late childhood or early adolescence. Thus, enamel is an archive of an individual's childhood (Grupe 1998). In contrast, bone tissue is remodeled over time. A turnover rate of about 5% per year is widely accepted for long bones of adult individuals (Martin et al. 2015). Accordingly, after 20 years, the bone is fully remodeled in any given region. Due to the ongoing remodeling of bone during lifetime, isotope analysis on bones allows to identify intra-life migration events of even adult individuals. As a consequence, individuals who migrated more than 20 years before death and subsequently consumed local food and water will most likely not be identified as primarily nonlocals at the site to which they migrated, but will reflect the local isotopic signature.

While univariate or bivariate analysis is still common when working with isotope data, by investigating the multi-isotope fingerprint of an individual we can obtain more detailed information about, e.g., diet, habitat, or provenance (Göhring et al. 2023). Accordingly, it is possible to identify primarily nonlocal individuals at a site who may have been invisible in the univariate or bivariate plots. The multidimensional approach, however, needs to be supported by machine learning approaches, such as cluster analysis (see below).

Machine learning approaches have been successfully applied to several multidimensional isotope datasets. In particular, so-called Gaussian Mixture Model (GMM) clustering has been proven to be a useful tool for investigating even very complex multi-isotope fingerprints in bioarchaeological studies (e.g., Göhring et al. 2023; Göhring et al. 2016; Grupe et al. 2020; Mauder et al. 2017; Tschetsch et al. 2020; Zacherl et al. 2024). Cluster analysis is a so-called unsupervised machine learning approach. Accordingly, the correct group label is not known beforehand. The cluster results need to be evaluated using metrics and also have to be verified by domain experts.

When performing cluster analysis on multi-isotope data, GMM cluster labels are not necessarily assigned with high confidence (here: clustering probability). This is especially the case for data points located between two or more clusters. Low clustering probabilities could indicate that an individual only partly fits into either of the clusters. Primarily nonlocal individuals may not necessarily be identified as mathematical outliers, as they may partially fit to the data due to, e.g., a mixed isotope signal in the case of intra-life migration. However, these individuals could be identified by a low clustering probability (Göhring et al. 2023; Grupe et al. 2020) and could be defined as so-called in-between instances (IBIs).

1.2 | In-Between Instances and Support Vector Machines

The IBI approach has only recently been proposed by Kazempour et al. (2024) and first applied to an isotope dataset by Göhring et al. (2024). IBIs can be considered as a third archetype besides inliers (class/cluster members) and outliers (anomalies that substantially deviate from class members). IBIs are located between two (or more) clusters and serve as connectors between these clusters. While they can be primarily regarded as outliers, since they do not belong to any class, IBIs can potentially exhibit characteristics shared by either cluster, reflecting a mixed signal of the clusters and thus distinguishing them from "classical" outliers, but also exhibit certain characteristics not explained by the clusters separately. Accordingly, they do not fully belong into neither of the clusters.

The task of identifying IBIs is related to the task of clustering and outlier detection. On the one hand, the identification of IBIs requires the differentiation of a certain IBI data point from the instances included in a cluster. On the other hand, in order to identify IBIs the similarity between an instance and the mixture of characteristics of two (or more) clusters has to be inferred.

In the present study, we examine if Support Vector machines (SVMs; Cortes and Vapnik 1995) can contribute to identify IBIs in multidimensional isotope data. In contrast to cluster analysis, SVMs are so-called supervised machine learning approaches. They are trained on a dataset containing labels indicating the class to which the data point belongs.

SVMs were originally introduced as a tool for binary classification. Binary classification aims at dividing a dataset into two classes (here: sites/regions) based on their inherent characteristics. Class separation is performed using a so-called decision boundary, with the two classes located on opposite sides of this boundary (Boser et al. 1992). SVMs search for an optimal hyperplane that performs best in differentiating between the two classes. More explicitly, the SVM identifies a separating hyperplane that maximizes the margin, i.e., the distance between the hyperplane and the nearest data points of each class. A larger margin makes the decision boundary less sensitive to variations in the training data, thereby reducing the risk of overfitting and improving generalization to unseen data.

The separating hyperplane is defined by support vectors (see Figure 1). Support vectors are the points closest to the decision boundary. The location of the support vectors determines the location of the separating hyperplane. The location of the optimal hyperplane as well as the identification of the support vectors are two equivalent problems that can be solved as a quadratic programming task, using the so-called kernel trick (for details see, e.g., Schölkopf 2001). Through the support vectors we have an implicit semantic of class borders while through the margin a geometric notion for the space between two classes is provided. The concept of the margin and the support vectors provides the means for IBI detection, discerning SVMs to other classification methods.

For our approach we choose three different kernel functions, namely, linear, Gaussian (rbf), and polynomial kernel. The

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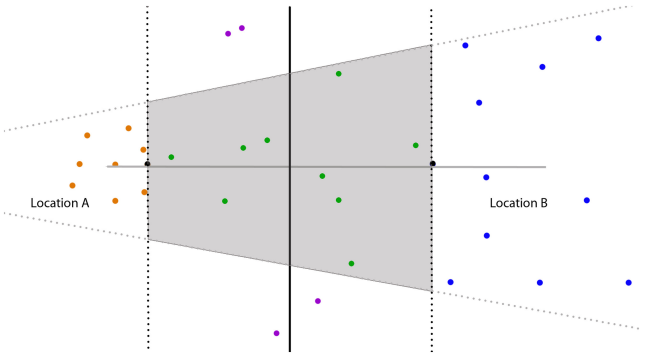


FIGURE 1 | IBI space between locations (Clusters) A and B defined by the distance to the separating boundary according to threshold T_1 (black dotted lines) and the distance to the cluster connecting vector (CCV) according to the dynamic threshold T_2 (gray dotted lines) (see text). The CCV can be understood as the vector spanned by the centers of two clusters (here: between Location A and Location B). The separating boundary (separating hyperplane) H is shown as bold vertical black line. Data points falling within the defined IBI space (highlighted in gray) are identified as IBIs (shown in green). Outliers falling outside the IBI space (shown in purple) are not identified as IBIs.

linear kernel represents the simplest kernel function. This kernel function is suitable for linearly separable datasets but fails for models with more complex patterns, hence differently shaped decision boundaries. The Gaussian kernel, also referred to as Radial Basis Function (rbf), implicitly maps the input features into an infinite-dimensional space. This kernel allows modeling more complex decision boundaries and thus capturing more complex data patterns. The polynomial kernel can be seen as a trade-off between linearity and nonlinearity. The degree of the polynomial allows setting a balance between model and decision boundary complexity and accuracy as well as effectiveness in separating the dataset in its respective classes (for details see, e.g., Boser et al. 1992).

Since a perfect separation into two classes is not guaranteed, within SVMs strict margins ("hard margins") and a relaxation of this strictness ("soft margin") can be distinguished. The level of strictness is controlled by parameter C, which regulates the error term introduced within the soft margin formulation of the SVM. This parameter controls the trade-off between maximizing the margin and minimizing the classification error on the training data. The choice of C is crucial to avoid overfitting. A high C value emphasizes the correct classification of all training samples. However, this can lead to more complex decision boundaries and thus lead to overfitting. Lower C values result in simpler and smoother decision boundaries with a larger margin between the classes. While more robust to overfitting, this could result in misclassification due to model underfitting and the model could fail to capture the patterns of the classes.

1.3 | Research Aims

The aim of the proposed SVM approach is to detect possible nonlocal individuals, identifiable as IBIs, as well as their possible sites or regions of origin, what needs to be further investigated using archaeological information (e.g., burial types, grave goods). Based on synthetic data, representing multidimensional isotope data, containing strontium and lead isotope ratios, we trained an SVM model capable of identifying IBIs between two clusters representing two sites or region. These IBIs could be understood as potential migrant candidates with mixed isotope signals. Moreover, since we expect the multi-isotope fingerprints of the IBIs to be the product of a linear mixing process over time due to bone remodeling (see Section 1.1), we aimed to approximate the years that passed since migration, to be understood as the years before death of the individual under study. In particular, we tested if our model is able to identify individuals who migrated almost 20 years ago. These individuals are most difficult to identify with common methods since they show isotope signatures very similar to the local signature of the site or region they migrated to. Finally, the SVM approach was applied on an archaeological isotope dataset of human cremated remains (Grupe et al. 2020). Some of the individuals in this dataset have been identified as (probably) nonlocal before based on their archaeological context (grave goods) in combination with a GMM clustering approach (Grupe et al. 2020).

2 | Material and Methods

2.1 | Synthetic Data

For an evaluation of the approach, synthetic data was used. A six-dimensional synthetic dataset, containing two reference clusters was generated referring to the isotope dataset of Grupe et al. (2020) (see Section 2.2). For a first experiment, the means and standard deviations observed by Grupe et al. (2020) were adjusted in order to create nonoverlapping clusters (Table S1). Random samples from Gaussian distributions with the presented mean and standard deviation were generated ($n = 200$ per cluster). The generated data were randomly split into training (75%) and test sets (25%).

Additionally, a set of IBIs was generated. Following our IBI definition (see Section 1.2), each IBI was associated with a start and end point within the clusters' distributions. Start and end points were generated based on the distributions of the reference clusters. For each set of start and end point, 20 evenly spaced points were generated lying in-between these points, representing the mixed data of an individual who migrated 0 to 20 years ago (see Figure S1). This set of generated IBIs does not only contain outliers but also points falling right in either of the reference cluster (e.g., migration 19 years ago). The number of IBIs generated for each experiment depended on the amount of cluster samples in the test set. We aimed to obtain a 50/50 split of positive and negative samples (migrant/no migrant) within our test set. Since the choice of the start and end points has a crucial effect on the model outcome, we run each experiment multiple times with different seeds (1 to 100) for generating the migrants (or IBIs). The metrics were calculated based on the mean performance of these runs. Different scenarios were tested:

1. The model was tested for varying number of samples in the reference clusters ($n = 25$, 50, 100, 200, 400, 1000, 2000, 20,000, 50,000, 100,000, 150,000, 200,000, 300,000, and

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500,000). Half of these data points were assigned to each of the two reference clusters, before splitting the data into training and test sets. The number of migrants was set as the multiple of 20 closest to the number of cluster samples in the test set. Accordingly, a test set close to 50% positive and 50% negatives was obtained.

2. In order to evaluate the impact of unbalanced and thus more realistic cluster sample sizes, the number of samples in the first cluster was set to 200, while the number of samples in the second cluster was flexible between 25 and 500,000 (see above).

3. Even differences in the standard deviation of the two reference clusters can have effects on our approach. To approximate the IBI (migrant) space, the threshold for the distance to the CCV (see Section 2.4) has to be adjusted to the position of the predicted sample in the orientation in which the clusters lie to each other. In addition, the reference clusters increasingly overlap with increasing standard deviations. A certain degree of overlap results in the collapse of the defined IBI space. In order to test how well our approach can adapt to clusters with different standard deviation and thus cluster overlap, the standard deviation of the created synthetic dataset was increased to generate the Gaussian distributed samples for all dimensions.

2.2 | Isotope Data

The approach obtained from the synthetic data (see above) was then also applied to real-world data. For a first test of our approach, we use the six-dimensional isotope dataset published by Grupe et al. (2020), containing strontium and lead ($^{87}Sr/^{86}Sr$, $^{208}Pb/^{204}Pb$, $^{207}Pb/^{204}Pb$, $^{206}Pb/^{204}Pb$, $^{208}Pb/^{207}Pb$, $^{206}Pb/^{207}Pb$) stable isotope data for 92 (mostly cremated) human remains of the inner Alpine Hallstatt culture in North Tyrol (older Iron Age; 650–450 BCE) and the early Fritzens–Sanzeno culture (younger Iron Age; 450—ca. 350 BCE) in North and South Tyrol.

In the mortuary ritual of the Hallstatt and Fritzens–Sanzeno cultures, the deceased were usually cremated and their cremated remains were buried in a grave with personal belongings (costume and jewelry) as well as ceramic food and drinking vessels. Some of these grave goods differ in appearance and style depending on their regional origin from the northern or southern Alpine region of the Hallstatt and Fritzens–Sanzeno cultures. They may have "migrated" with their owners, been imported, or even inspired imitation in the other region.

The Fritzens–Sanzeno culture developed in South Tyrol and Trento. The archaeological question is whether its spread into the northern Alps was connected with the immigration of larger population groups from South Tyrol into the North Tyrolean Inn Valley and neighboring areas, or whether increased contacts in the context of the exchange of goods and technical know-how on the Inn–Eisack–Etsch–Brenner passage also led to a similar cultural development (Metzner-Nebelsick et al. 2017).

The assessment criterion or indicator for this question of the extent of mobility is grave goods that were characteristic for the northern or southern Alpine area of the Fritzens–Sanzeno culture. For this reason, 92 graves from important sites in North and South Tyrol were selected with variously composed grave goods. The research conducted by Grupe et al. (2020) on the human remains supported the hypothesis that the spread of the Fritzens–Sanzeno culture into the area north of the Alps was the result of increased contacts, with individual migrants, rather than whole populations. Several individuals were identified to be nonlocal to the region (Grupe et al. 2020).

The dataset (see also Table S2) includes isotope data for burials from the Inn Valley (Wörgl [$n = 11$], Kundl [$n = 48$]) and South Tyrol (Moritzing [n = 11], Pfatten [n = 4], Latsch [$n = 16$], Brixen [$n = 2$]; see Figure 2).

2.3 | Model Assumptions

Two main assumptions are made: First of all, we assume a constant bone remodeling rate of 5% per year (see Section 1.1). Assuming a constant remodeling rate is required to calculate the isotope signature of the IBIs, labeled as migrants, in order to train our model. Under normal conditions, some variability in the bone remodeling rate within an individual's organism or between different individuals can occur. Nonadult individuals show increased remodeling rates. Physiological parameters can have an impact on bone remodeling as well (Stout et al. 2019). Nevertheless, for training purposes a linear remodeling is assumed (see Figure S1), certainly simplifying the underlying remodeling processes.

Moreover, we assume that the reference clusters, representing the isotope signature of two sites or regions, are Gaussian distributed. Biological measures, including isotope data, should usually be normally distributed. The suitability of a Gaussian distribution assumption for isotope data has been demonstrated in previous studies (see Introduction). Accordingly, this can be seen as a reasonable assumption.

2.4 | SVM Approach

Migrant candidates are expected to fall into the IBI space (see Section 1.2). Accordingly, the latter needs to be defined. Based on the assumptions (see Section 2.3), our approach can be divided into two steps: In the first step, the IBI space is bounded using the hyperplane provided by the SVM (see Figure 1). In the second step, the IBI space is constrained using the Cluster Connecting Vector (CCV; see Figure 1). The implementation is available at Github (https://github.com/janb15/There-may-be-migrants-in-between-Support-Vector-Machines-and-cluster-in-betweenness). In the following, we shortly summarize the two steps:

Step 1: Confinement based on the SVM model

The SVM approach provides us with the support vectors that are specific points of our training set that define the position of the decision boundary. Moreover, the model provides the weights that are the coefficients of the vector, determining the coordinates of the decision boundary in the kernel space, as well as the

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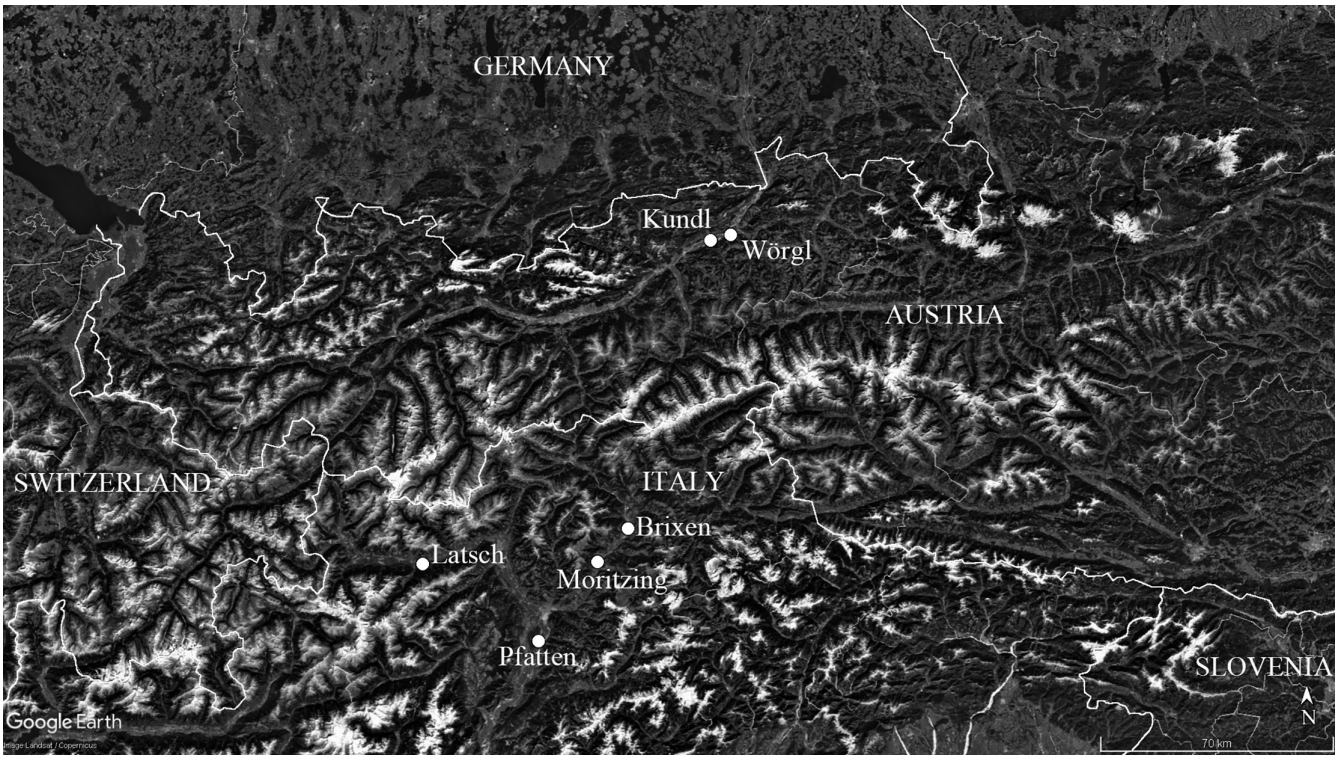


FIGURE 2 | Map indicating the Fritzens–Sanzeno burial sites investigated by Grupe et al. (2020). Google Earth Pro, Google 2024, https://earth.google.com/. Alpine map: 01/01/2021, 48.895869° N, 11.921589° E, eye altitude 425.40 km.

decision function. The latter is used to classify new data points based on the established decision boundary.

The threshold (T_1) for the distance to the separating boundary has an impact on how sharply the IBI space is defined. A larger threshold results in the identification of cluster points as IBIs (false positives), while a lower threshold may lead to the omission of some IBIs and thus the increase of false negatives. The aim of setting a threshold is to exclude data points from within the reference cluster. Since the clusters are Gaussian distributed, we assume that the distances D are also Gaussian distributed. This is controlled by evaluating Q–Q plots as well as the kurtosis and skewness of the distribution.

After obtaining the set of distances D, the mean and standard deviation of D are calculated. We test the performance of our approach with different thresholds based on a multiplier of the standard deviation of D.

Step 2: Confinement based on the CCV

To further restrict the IBI space, the IBI space was confined by the CCV. The CCV can be understood as the vector spanned by the centers of two clusters A and B, thus orthogonally to the separating boundary. This allows distinguishing between data points belonging to one of the reference clusters and IBIs (see Figure 1; gray line orthogonal to H). By calculating the distance to the CCV, it is possible to measure how far a data point deviates from both clusters in this particular orientation. A dynamic threshold T_2 for this distance can be approximated by examining the distances D of the data points of each cluster to the CCV. Similar to T_1 (see above), we expect D to be Gaussian distributed. Analogous for the distance to the separating boundary (see above), we calculate the mean and standard deviation. Different multipliers are tested within the experiments.

For a model being able to handle clusters of highly different spread (see Figure 1), a dynamic threshold is required. To address this requirement, one cluster is chosen and vector x_A between its center and data point x is calculated. Then, the projection $\mathrm{proj}_C x_A$ of vector x_A onto the CCV is calculated with

$$\mathrm{proj}_C x_A = \frac{x_A \cdot \mathrm{CCV}}{C \cdot \mathrm{CCV}} \mathrm{CCV}. \quad (1)$$

The distance traveled along the CCV compared to the full length of the CCV is measured as the fraction (*frac*) of the length of the projected vector and the CCV:

$$\mathrm{frac} = \frac{\|\mathrm{proj}_{\mathrm{CCV}} x_A\|}{\|\mathrm{CCV}\|} \quad (2)$$

Finally, the thresholds for Clusters A and B, T_A and T_B, are used to calculate the new dynamic threshold with

$$\mathrm{threshold} = T_A + \left(\left(T_B - T_A\right) * \mathrm{frac}\right). \quad (3)$$

If frac is below 1, the data point x falls between the cluster centers and can be defined as an IBI. The threshold is then set according to the calculations. If frac is larger than 1, the data point x lies behind the (closest) cluster center and thus outside our IBI space. Accordingly, the threshold is set to $-\infty$. This value cannot be satisfied within the threshold check, and thus, the data point will not be labeled as an IBI.

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2.5 | Approximation of the Time Since Migration

The migration of an individual (data point) from one location to another location initiates a linear mixing process of the isotope signatures of both locations (see Section 2.1). Cluster representatives can be used as proxies of the start and end points.

Assuming an identified migrant candidate x, originating from Site A with cluster mean μ_A. The mean of the cluster representing Site B, where the individual was buried, is denoted as μ_B. The CCV can be calculated according to $CCV = \mu_A - \mu_B$. Then, the years since migration can be estimated as follows: x_A is the vector between the location of the sample x and the center of its origin cluster μ_A. The fraction of the projection of this vector onto the CCV ($proj_{CCV} x_A$) is calculated analogous to Equation (1). The full CCV describes the fraction how far the mixing process has progressed, given starting point μ_A and end point μ_B (analogous to Equation 2). This fraction has to be rescaled by 20 years, assuming a complete remodeling after 20 years in order to approximate the years since migration:

$$\text{years since migration} = 20 * \text{frac} \tag{4}$$

with frac calculated according to Equation (2).

3 | Results

3.1 | Evaluation of the SVM Approach

When performing the model with different numbers of samples in the reference clusters ($n = 25$ to $n = 250{,}000$), we observe satisfying performance across all configurations as illustrated in Figure S2. According to the scores, performance was best with 100 samples per cluster. Low sample sizes ($n = 25$ or 50) produce worse results than larger ones. With a sample size of 200 or higher per cluster we see a consistently good performance (Figure S2).

In more realistic scenarios, the sample size between the reference clusters differs. Thus, we tested a constant sample size of 200 for one of the clusters against a sample size varying between 25 and 250,000 for the second cluster. All configurations result in high accuracy, precision, and F1 scores, while the recall indicates some false negatives (0.83–0.85; Figure S3).

Figure S4 illustrates the results for the evaluation if our algorithm can identify IBIs that are close to the reference clusters, thus individuals who migrated almost 20 years before death or shortly before death. For a configuration of $T_1 = 2$, we observe a distinct increase of all metrics, besides the constantly high precision, for the range of 5–8 years since migration. The performance then reaches a peak (9–12 years) until it declines for individuals who migrated 16–20 years before death, albeit slower than the increase observable at 5–8 years. If T_1 is set to 1.5, similar results are observed with a constantly good performance between 5 and 16 years, followed by a decline in performance for individuals who migrated 16–20 years before death.

When investigating the performance for every year separately (see Figure S5), we observe a distinct increase in all metrics within the first five years, followed by constantly high scores until candidates who migrate 17 years before death. Then, the trend reverses and the scores begin to decrease, comparable to the results shown in Figure S4. The experiment's MSE and RMSE scores are 0.996 and 0.998, respectively, indicating a high level of accuracy.

Finally, the impact of spread differences between clusters was investigated. Figure S6 illustrates the performance of our algorithm for a linear kernel. Generally, the performance is independent from the choice of C regarding the general trend of all metrics. Starting with high scores for clusters exhibiting the same standard deviation, all metrics decrease with increasing multipliers for the standard deviation, thus increasing levels of overlap. Similar results are observable for the rbf kernel (Figure S7).

The decline in the performance of the algorithm is related to the number of overlapping dimensions, as shown in Figure S8. For each multiplier, the number of overlapping dimensions between the clusters was calculated. At a multiplier of 1.6, the clusters begin to overlap in at least two dimensions, and at multipliers above 2.8, they overlap in all six dimensions examined. However, even though the clusters overlap within multiple dimensions, for all configurations the SVM algorithm was able to separate the classes without misclassification on the test set.

3.2 | Application on Fritzens–Sanzeno Isotope Dataset

Table S3 shows the ten best performing models resulting from the parameter tuning process on the Fritzens–Sanzeno dataset (Grupe et al. 2020). All these models use a Gaussian (rbf) kernel, appearing to be most effective at identifying the underlying patterns in the data. This comes to no surprise as these underlying processes are expected to be Gaussian distributed (see Section 2.3). All models use a comparable C value, ranging between 2.656E8 and 1E10. A substantial penalty for misclassification is thus required in order to capture the data structure. This, however, results in more complex decision boundaries. In order to mitigate overfitting, we employed 5-fold cross validation and computed the F1 score as the average of all five folds. The F1 score for all models is consistently high with 0.97. This indicates that the performance of the model is robust to variations in C, provided that C remains sufficiently high. The F1 score indicates that all 10 models presented in Table S3 accurately capture the structure of the isotope dataset.

In order to obtain more variable models and thus decision boundary structures, we also selected the two best performing models for each of the three investigated kernels (linear, rbf, polynomial) as shown in Table S4. The models with rbf kernels represent the two best performing models of Table S3. However, the F1 scores for the best models using the polynomial kernel (0.96) or linear kernel (0.92) are also capable to separate the reference clusters of the inner Alpine and South Tyrolean burial sites.

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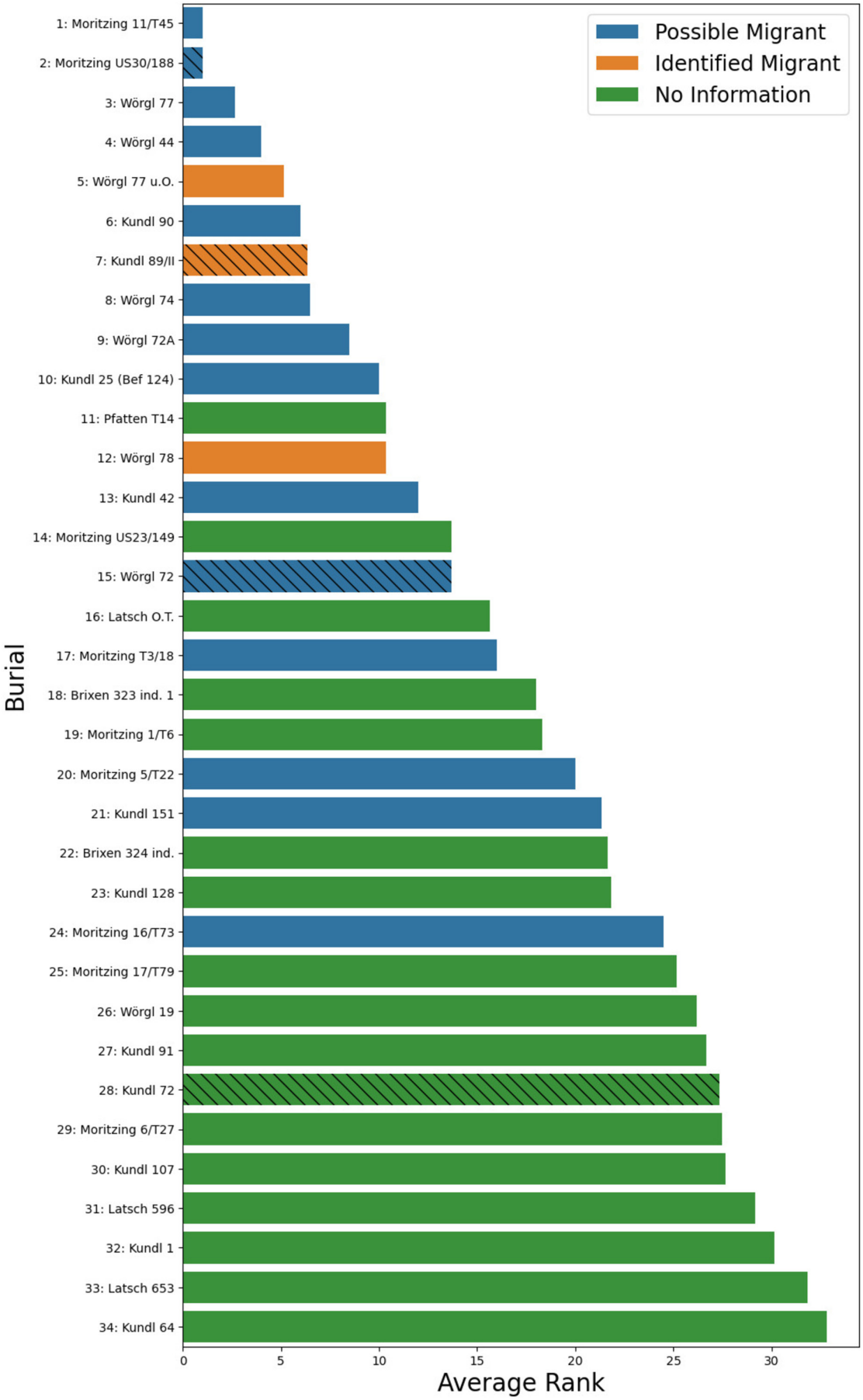


FIGURE 3 | Legend on next page.

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FIGURE 3 | Ranking of identified migrant candidates of the Fritzens–Sanzeno isotope dataset (Grupe et al. 2020) using the two best performing models per kernel (see Table S4). Burials with archaeologically conspicuous burial context (deviating burial rites) are shaded.

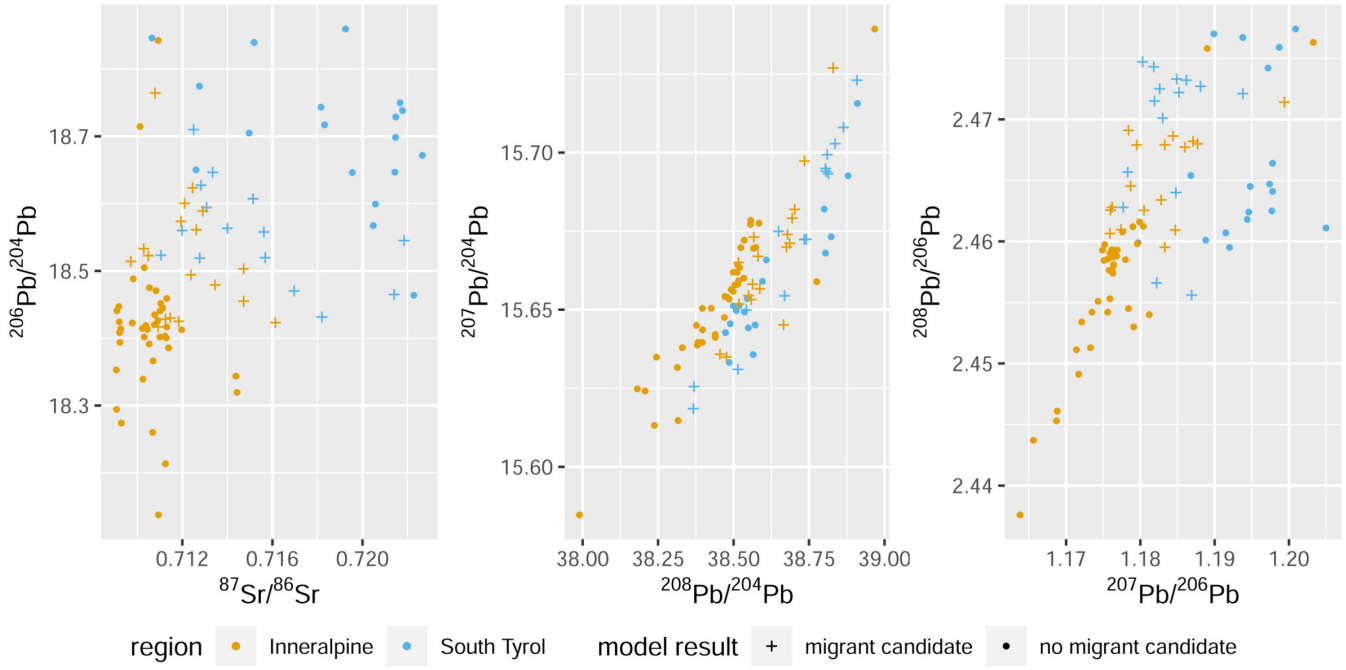


FIGURE 4 | Visualization of the isotope fingerprint of migrant candidates identified within the Fritzens–Sanzeno isotope dataset (Grupe et al. 2020) using the two best performing models per kernel.

The results of the migrant detection for the Fritzens–Sanzeno dataset using the ten best performing models and the two best performing models per kernel are illustrated in Figures 3 and S9, respectively. As visible in Figure 4, the identified migrant candidates exhibit isotope fingerprints representing a variable mixture of the isotope signatures of the reference clusters of the inner Alpine region and South Tyrol, respectively. For better readability, the ranking (Figures 3 and S9) only represents those individuals identified as migrant candidates in at least one configuration. The model is most confident in samples with a low rank being mixed isotope signals of the reference clusters thus migrant candidates. With increasing average rank, the confidence decreases. It has to be noted that a higher average rank could also be due to a migration that happened long ago. The ranking should, however, not be understood as any kind of probability measure. We rather advise to investigate the identified migrant candidates in more detail (Figures 3 and S9), also with respect to the archaeological context of the burials. Individuals with an average rank of 30.9 or 34.0 within the 10 best models and the two best kernels per models, respectively, were not included. These are most likely no migrant candidates.

Besides three individuals already identified as migrants by Grupe et al. (2020), partly also based on nonlocal grave goods, our model suggests a nonlocal origin for 14 individuals identified as possible migrants by Grupe et al. (2020) (statistical outlier, overlapping cluster region). Moreover, we identify additional 14 and 16 individuals, respectively, not identified as migrants before. For three of these individuals (Kundl 72, Kundl 89/II, Wörgl 72; two best performing models per kernel), the archaeological context could match a nonlocal origin. For the others, the burial context does not allow a detailed evaluation.

Our algorithm does not identify the individuals Kundl 40, Kundl 89/I, or Kundl 160 as migrant candidates (see Figures 3 and S9), while these individuals have been identified as nonlocal by representing their own GMM cluster (Grupe et al. 2020), also peculiar from their archaeological context. This is discussed below.

3.3 | Predicted Years Since Migration

Tables 1 and S5 show the approximated years after migration (years before death) for the identified migrant candidates using the two best performing models per kernel and the 10 best performing models, respectively (see corresponding Figures 3 and S9). It is important to mention that these approximated results shall not be understood as precise measures. They shall rather give a rough approximation, certainly dependent on the data basis of the reference clusters. For the provided ranges, an RMSE of 0.996 years (see above) is applied, rounded to whole numbers.

With increasing rank of the migrant candidates, the linear regression line and thus the number of years since migration increase. Not surprisingly, higher ranked individuals tend to have migrated more recently and are therefore easier to identify. The confidence of the migrant prediction is higher for the higher ranks as they are found to be migrant candidates with a smaller IBI space than the lower ranked sample. The confidence interval suggests reasonable certainty about the trend. However,

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TABLE 1 | Predicted years since migration (mathematically exact value as well as plausible range, based on the accuracy of prediction of 0.996 years and rounded to whole numbers) for the migrant candidates identified within the Fritzens–Sanzeno isotope dataset (Grupe et al. 2020) using the two best performing models per kernel (see Figure 3), sorted by years since migration.

Site	Burial	Predicted years since migration	Predicted years since migration (range)
Kundl	89/II	3.52	2–5
Wörgl	72A	7.18	6–9
Moritzing	US30/188	7.84	6–9
Wörgl	77 u.o.	8.07	7–10
Kundl	90	8.74	7–10
Wörgl	44	9.32	8–11
Moritzing	T3/18	9.61	8–11
Kundl	42	9.83	8–11
Moritzing	11/T45	11.44	10–13
Latsch	O.T.	11.59	10–13
Brixen	324 ind.	11.88	10–13
Wörgl	77	11.91	10–13
Wörgl	78	12.22	11–14
Wörgl	74	12.49	11–14
Brixen	323 ind. 1	12.89	11–14
Kundl	25 (Bef 124)	12.91	11–14
Kundl	151	13.16	12–15
Wörgl	72	13.70	12–15
Moritzing	5/T22	14.75	13–16
Latsch	653	15.46	14–17
Moritzing	16/T73	15.56	14–17
Moritzing	US23/149	15.74	14–17
Moritzing	1/T6	16.99	15–18
Kundl	107	17.53	16–19
Pfatten	T14	17.62	16–19
Kundl	91	17.92	16–19
Moritzing	17/T79	18.14	17–20
Kundl	128	18.21	17–20
Moritzing	6/T27	18.24	17–20
Kundl	64	18.77	17–20
Latsch	596	19.05	18–21
Kundl	1	19.22	18–21
Wörgl	19	19.80	18–21
Kundl	72	20.10	19–22

there is a noticeable variability among the predicted years since migration for the investigated individuals.

Table 1 gives an overview on the approximate years since migration for the identified migrant candidates (two best performing models per kernel), ranging between about 3 years (Kundl 89/II) and about 19 or even 20 years (Latsch 596, Kundl 1, Wörgl 19, Kundl 72). Individuals with predicted years since migration close to 20 years should be treated with caution since the similarity between the individuals' isotopic fingerprint and the respective reference cluster is very high and could also reflect normal variability rather than a nonlocal origin. Nevertheless, an increase in T_{1m}, and a consequent shift of the threshold of the IBI space in the direction of the clusters, is not advisable. Since our approach allows identifying even individuals who migrated almost 20 years before death, it can be assumed that we identified all reasonable possible migrant candidates.

4 | Discussion

4.1 | Evaluation of the Approach

Our experiments on the synthetic datasets demonstrate that the approach is generally resistant to varying sample sizes. While low sample sizes of 25 or 50 cause some instability in the results, the performance remains high across all tested sample sizes. Related to the sample size, we also observe more variable results in clusters with low density. Nevertheless, the performance remains robust across all configurations.

Individuals who migrated many years before they died (up to 20 years before death) are most difficult to identify due to the similarity of their isotope fingerprint with the local isotope signal. Individuals who migrated 19 years or more before death cannot securely be identified by our approach, while it is possible to detect migrants who migrated up to 18 years before death. 19 or even 20 years after migration, the individual's isotope fingerprints cannot be distinguished from the reference cluster. Our algorithm allows using the parameter T_{1m} to choose a tradeoff between more secure labeling, with a lower number of false positives, and higher sensitivity, with a lower number of false negatives.

If the spread of one of the reference clusters increases, the performance of our algorithm deteriorates. This can be explained by the increasing overlap of the reference clusters and a subsequent collapse of the IBI space. If the isotopic signature of two sites is too similar or even identical, it is no longer possible to identify potential migrants with an isotopic approach, regardless of whether machine learning is used or not. Overall, however, our evaluation confirms that our approach is capable of identifying migrants within an isotopic dataset.

4.2 | Migrant Candidates and Migration Events

Our approach allows identifying 34 (two best performing models per kernel) and 31 (ten best performing models) potentially nonlocal individuals in a multidimensional isotope dataset, as demonstrated for the dataset from the Alpine region.

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Some individuals of the Fritzens–Sanzeno isotope dataset have already been identified as nonlocal using a GMM clustering approach, partly also suggested by grave goods indicating nonlocal origin (Grupe et al. 2020). Others were identified as possible migrants. These were mathematical outliers or they were found in overlapping regions of GMM clusters, thus reflecting a low clustering probability (Grupe et al. 2020). Our approach also identifies three individuals with high confidence (far ahead in the ranking; Figures 3 and S9) who have already been identified as potential migrants by Grupe et al. (2020), namely, Wörgl 77, Kundl 89/II, Wörgl 78. For several of the possible migrants ($n = 14$; see Grupe et al. 2020), our approach allows to be more confident in identifying these individuals as primarily nonlocal (Figures 3 and S9). A high number of individuals cannot be identified as potentially nonlocal based on the archaeological burial context or the GMM approach. However, our algorithm allows adding another 17 (two best performing models per kernel) and 11 (ten best performing models) individuals to the list of migrant candidates, respectively (Figures 3 and S9). This includes also individuals who were conspicuous with respect to their archaeological context. The burials Kundl 72, Kundl 89/II, and Wörgl 72 contained ceramics, pointing to a connection to the South Alpine region, Southwest Germany, and potentially the Alpine foothills, respectively. In contrast to typical Fritzens–Sanzeno burial rites, no grave goods were found within the burial Moritzing US30/188. All these individuals were identified as IBIs and thus potential migrants by our approach.

However, as already mentioned above, our model does not identify three individuals from Kundl (Kundl 40, Kundl 89/I, Kundl 160) identified as nonlocal by the GMM approach, including a burial (Kundl 160) containing an amulet of potential South Alpine origin. These individuals were previously grouped into a separate cluster (Grupe et al. 2020). We assume that these individuals may have migrated from a different location than tested in our model. This is also in accordance with Grupe et al. (2020), who proposed that these individuals were not grouped together due to a common origin but rather due to their difference from the other data points. While ceramic of Kundl 89/I points to southwestern Germany, the amulet of Kundl 160 points to the south, while the grave of Kundl 40 contained unique ceramic (Grupe et al. 2020). An origin from a location different from the investigated region south of the Alps seems plausible and can explain the failure of our model to identify these individuals. It is important to highlight that the choice of the reference clusters, used to test specific potential regions of origin, has a clear impact on the model results. These regions of origin have to be selected in close collaboration with the domain experts from archaeology.

As already mentioned before, the approximation of the years since migration, representing years before death, is not aimed to be an accurate measure. Since we only have some representatives for both reference clusters—instead of the whole local population—this measure is, of course, also dependent on the respective reference clusters. Accordingly, while provided as exact measures, resulting from the underlying mathematical approach, we rather recommend considering the range calculated based on the RMSE of 0.996 years (see Tables 1 and S5).

Moreover, it is important to mention that the two individuals Wörgl 77 and Wörgl 77 u. O., identified as migrant candidates by our approach, were quite young (max. Juvenile; see Table S2). Younger individuals have a faster bone turnover rate than adults. Accordingly, the approximated years since migration (Wörgl 77: 10–13 years; Wörgl 77 u. O.: 7–10 years; see Table 1) are probably overestimated in the case of these individuals. Assuming a larger bone turnover rate for these two individuals of, e.g., 30%, expected for fast growing children, would result in an approximated migratory event that took place much later in life, namely, about 1.98 (0–3 years; Wörgl 77) and 1.34 years (0–3 years; Wörgl 77 u. O.) before death.

4.3 | Limitations

The presented approach is limited to reference regions (clusters) exhibiting a nonoverlapping isotopic signature. Otherwise, the IBI space would be too small to be meaningful or could not even be defined. This is a general limitation of provenance analysis based on isotope data. Individuals originating from another site that showed the same local isotope signature as the site the individuals migrated to cannot be identified as nonlocal. In general, while we can identify individuals with a somewhat different isotope fingerprint as nonlocals, we can never be sure that all the inconspicuous individuals of the site (or region) were actually locals. Usually, it is assumed that the majority of individual at a site represents the local isotope signature. This is certainly not the case for newly established sites, after a (forced) relocation, a catastrophe, etc.

Our approach uses the simplifying assumption that bone remodeling and respective mixing of isotope signatures occurs in a very regular and linear way. While bone in adult individuals, on average, remodels about 5% per year (see Section 1.1), this can vary both intra-individually (different bones) and inter-individually. In addition, the bone remodeling rate is higher in children. Our approach, however, allows an adjustment for the expected remodeling time (by adjusting Equation (4)). Thus, if isotope analysis was conducted on, e.g., ribs with a faster remodeling rate (Cox and Sealy 1997), the presented approach can also be used by training the data with another set of migrants that fit the different remodeling rate. Moreover, it is important to mention that our proposed IBI approach itself does not use the remodeling rate. The latter is only important for the approximation of the years since migration. Variability due to, e.g., inter-individual differences can be incorporated by training several models with IBIs generated based on different turnover rates.

In addition, we consider migratory events between two sites A and B. The two clusters approach ignores migratory events from another site C. The model checks for potential migrants from site A to site B (or the other way around). Similarly, it can check for potential migrants from site C to A or B, respectively. However, this cannot be conducted in parallel. Archaeological background information (e.g., grave goods, burial rites) can help to identify potential regions of origins, which should be checked by the model.

Moreover, the algorithm is based on the assumption that the isotope data follows a Gaussian distribution. This was shown to be

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a meaningful assumption for different isotope datasets, also including datasets with a quite small number of samples (Göhring et al. 2023; Göhring et al. 2016; Grupe et al. 2020; Zacherl et al. 2024). However, depending on the collection of samples for isotope analysis, care must be taken. It is important that the reference clusters represent samples of the "normal" population of the site or region under study, not only burials with peculiar archaeological contexts or, in particular, individuals with nonlocal burial goods. This is also a general advice for isotope analyses rather than restricted to our approach. The reference clusters resulting from such a sampling procedure are most likely not Gaussian distributed and will not allow the prediction of nonlocals using our algorithm.

Our IBI definition implies that each IBI is an outlier. Indeed, individuals close to the reference cluster, not identified as outliers, are most difficult to be identified as migrants. However, not all nonlocal individuals are necessarily identified as outliers with respect to their isotopic signature. This is especially the case for individuals who migrated more than 20 years before death, usually representing the local isotopic signature. This is, of course, not only a problem of our approach but of all isotope studies working with (remodeling) bone samples. Isotope analysis on enamel, reflecting the isotope signature during early childhood, is often used instead. However, enamel data do not allow the identification of migration during, e.g., adulthood. In addition, our approach is only possible under the assumption of remodeling resulting in IBIs.

5 | Conclusion

This study provides a novel algorithm for provenance analysis in multi-isotope datasets. It provides a fully data-driven approach that can be used to obtain a ranking of migrant candidates or to make an individual decision on the trade-off between capturing (very) early migrants that are difficult to identify and more confident labeling by manually setting the parameters of the algorithm.

Using both synthetic data and real-world archaeological isotope data, this study demonstrates the potential of SVMs combined with the concept of cluster in-betweenness for identifying migrants within multi-isotope datasets.

It is important to emphasize that the model identifies migrant candidates with varying confidence, depending on the numbers of years since migration. These candidates must be further analyzed archaeologically with regard to burial ritual, burial custom, and grave goods.

While testing the approach on synthetic data and on isotope data measured in archaeological samples obtained convincing results, indicating a well-performing and robust algorithm, the approach needs to be further tested on additional datasets.

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Data Availability Statement

The data supporting the results of our study are available in the Supporting Information of this article. The Python script, including the synthetic data, is available at Github (https://github.com/janb15/There-may-be-migrants-in-between-Support-Vector-Machines-and-cluster-in-betweenness).

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Exemplary IBI generation for migrants from Site A to Site B over the course of 20 years, represented as a bivariate plot showing only two isotope systems ($^{87}Sr/^{86}Sr$, $^{208}Pb/^{204}Pb$; synthetic data). **Figure S2:** Evaluation of the effect of the number of samples in the reference Clusters 1 and 2, with $n_A = n_B$ on the model outcome. **Figure S3:** Evaluation of the effect of unbalanced reference clusters with different number of samples in Cluster B and a constant number of 200 samples in Cluster A on the model outcome. **Figure S4:** Evaluation of the impact of the migration progress by groups of 4 years on the model outcome for (A) $T_1 = 2$ and (B) $T_1 = 1.5$. **Figure S5:** Evaluation of the impact of the migration progress for individual years (since migration) on the model outcome for (A) $T_1 = 2$ and (B) $T_1 = 1.5$. **Figure S6:** Evaluation of the impact of spread differences and cluster overlap on the model outcome using a linear kernel and parameter (a) $C = 10E10$ and (b) $C = 1$, respectively. **Figure S7:** Evaluation of the impact of spread differences and cluster overlap on the model outcome using an rbf kernel and parameter (a) $C = 10E10$ and (b) $C = 1$, respectively. **Figure S8:** Relationship between the number of overlapping dimensions and increasing standard deviation (sd) affecting the model outcome (see text). **Figure S9:** Ranking of identified migrant candidates of the Fritzens–Sanzeno isotope dataset (Grupe et al. 2020) using the 10 best performing models (see Table S3). Burials with archaeologically conspicuous burial context (deviating burial rites) are shaded.

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Table S1: Mean (μ) and standard deviation (sd) of Gaussian distributions for reference Clusters 1 and 2 of the synthetic dataset adapted from Grupe et al. (2020). **Table S2:** Isotope dataset published by Grupe et al. (2020) including information on site, burial ID, age-at-death (juv. = juvenile), sex (m = male, f = female, nd = indet.), strontium, and lead isotope ratios measured in the bones, cluster assignment for clustering the whole dataset (Sr + Pb) and for clustering with lead isotopes only (Pb only), including clustering probability p for either of the clusters. **Table S3:** Settings (kernel, C, gamma) for the 10 best performing models (with average F1 scores) in the experimental setup. **Table S4:** Settings (C, gamma) for the two best performing models (see F1 score) for each kernel (rbf, polynomial, linear) in the experimental setup. **Table S5:** Predicted years since migration (exact value as well as range, based on the accuracy of prediction of 0.996 years, rounded to whole numbers) for the migrant candidates identified within the Fritzens–Sanzeno isotope dataset (Grupe et al. 2020) using the 10 best performing models (see Figure S9), sorted by years since migration.

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